

CBSE BOARD EXAMINATION 2024

MATHEMATICS (STANDARD)

Class 10th — Solved Paper with Answers

Maximum Marks:	80
Time Allowed:	3 Hours
Date of Exam:	March 11, 2024
Total Questions:	38 (All Compulsory)
Sections:	A (MCQs), B (VSA), C (SA), D (LA), E (Case Study)

GENERAL INSTRUCTIONS:

- (i) This question paper contains 38 questions. All questions are compulsory.
- (ii) This question paper is divided into five sections — A, B, C, D and E.
- (iii) Section A: Q1–Q18 are MCQs (1 mark each); Q19–Q20 are Assertion-Reason (1 mark each).
- (iv) Section B: Q21–Q25 are Very Short Answer type questions (2 marks each).
- (v) Section C: Q26–Q31 are Short Answer type questions (3 marks each).
- (vi) Section D: Q32–Q35 are Long Answer type questions (5 marks each).
- (vii) Section E: Q36–Q38 are Case Study based questions (4 marks each).
- (viii) There is no overall choice. Internal choice is provided in some questions.
- (ix) Draw neat diagrams wherever required. Take $\pi = 22/7$ wherever required.
- (x) Use of calculators is NOT allowed.

SECTION A — MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q1. The value of k for which the system of equations $3x - y + 8 = 0$ and $6x - ky + 16 = 0$ has infinitely many solutions, is

- (A) -2
- (B) 2
- (C) $1/2$
- (D) $-1/2$

Answer: (B)

Solution: For infinite solutions: $a_1/a_2 = b_1/b_2 = c_1/c_2$. $3/6 = -1/(-k) \rightarrow 1/2 = 1/k \rightarrow k = 2$.

Q2. Point P divides the line segment joining $A(4, -5)$ and $B(1, 2)$ in the ratio $5:2$. Coordinates of P are

- (A) $(5/2, -3/2)$
- (B) $(11/7, 0)$
- (C) $(13/7, 0)$
- (D) $(0, 13/7)$

Answer: (C)

Solution: $x = (5 \times 1 + 2 \times 4)/(5+2) = 13/7$, $y = (5 \times 2 + 2 \times (-5))/7 = 0$. So $P = (13/7, 0)$.

Q3. The common difference of an A.P. in which $a_{14} - a_{10} = 48$, is

- (A) 12
- (B) 16
- (C) -12
- (D) -16

Answer: (A)

Solution: $a_{14} - a_{10} = (a + 14d) - (a + 10d) = 4d = 48 \rightarrow d = 12$.

Q4. The quadratic equation $x^2 + x + 1 = 0$ has roots

- (A) real and equal
- (B) irrational
- (C) real and distinct
- (D) not real

Answer: (D)

Solution: $D = b^2 - 4ac = 1 - 4 = -3 < 0$. Since $D < 0$, roots are not real (imaginary).

Q5. If $\text{HCF}(2520, 6600) = 40$ and $\text{LCM}(2520, 6600) = 252 \times k$, then k is

- (A) 1650
- (B) 1600
- (C) 165
- (D) 1625

Answer: (C)

Solution: $\text{HCF} \times \text{LCM} = \text{Product} \rightarrow 40 \times (252 \times k) = 2520 \times 6600 \rightarrow k = (2520 \times 6600)/(40 \times 252) = 165$.

Q6. In $\triangle ABC$, $DE \parallel BC$. If $AD = 5$ cm, $DB = 2.5$ cm and $BC = 12$ cm, then DE is

- (A) 10 cm
- (B) 6 cm
- (C) 8 cm
- (D) 7.5 cm

Answer: (C)

Solution: By BPT: $AD/AB = DE/BC \rightarrow 5/7.5 = DE/12 \rightarrow DE = (5 \times 12)/7.5 = 8$ cm.

Q7. If $\sin \theta = \cos \theta$ ($0^\circ < \theta < 90^\circ$), then value of $\sec \theta \cdot \sin \theta$ is

- (A) $1/\sqrt{2}$
- (B) $\sqrt{2}$
- (C) 1
- (D) 0

Answer: (C)

Solution: $\sin \theta = \cos \theta \rightarrow \tan \theta = 1 \rightarrow \theta = 45^\circ$. $\sec 45^\circ \cdot \sin 45^\circ = \sqrt{2} \times 1/\sqrt{2} = 1$.

Q8. Two dice are rolled together. The probability of getting sum more than 10 is

- (A) $1/9$
- (B) $1/6$
- (C) $1/12$
- (D) $5/36$

Answer: (C)

Solution: Sum > 10 means sum = 11 or 12. Favourable: (5,6), (6,5), (6,6) = 3 outcomes. $P = 3/36 = 1/12$.

Q9. In the figure, PT is tangent to a circle with centre O. Chord PQ subtends 65° at centre. $\angle QPT$ is

- (A) 25°
- (B) 32.5°
- (C) 65°
- (D) 90°

Answer: (B)

Solution: $\angle POQ = 65^\circ$. In $\triangle POQ$, $OP = OQ$ (radii). $\angle OPQ = \angle OQP = (180-65)/2 = 57.5^\circ$. $\angle OPT = 90^\circ$ (tangent). So $\angle QPT = 90^\circ - 57.5^\circ = 32.5^\circ$.

Q10. If α and β are zeroes of $2x^2 - 9x + 5$, then $\alpha^2 + \beta^2$ is

- (A) $1/4$
- (B) $61/4$
- (C) 1
- (D) $71/4$

Answer: (B)

Solution: $\alpha + \beta = 9/2$, $\alpha\beta = 5/2$. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 81/4 - 5 = 81/4 - 20/4 = 61/4$.

Q11. Value of k for which $x = 2$ is a solution of $5x^2 - 4x + (2+k) = 0$, is

- (A) 10
- (B) -10
- (C) 14
- (D) -14

Answer: (D)

Solution: Put $x = 2$: $5(4) - 4(2) + 2 + k = 0 \rightarrow 20 - 8 + 2 + k = 0 \rightarrow 14 + k = 0 \rightarrow k = -14$.

Q12. The sum of first 200 natural numbers is

- (A) 2010
- (B) 2000
- (C) 20100
- (D) 21000

Answer: (C)

Solution: $S_{\blacksquare\blacksquare\blacksquare\blacksquare} = 200 \times 201/2 = 20100$.

Q13. A hollow cylinder of length 14 cm has outer and inner radii difference 1 cm. If volume of metal is 176 cm³, the outer radius is

- (A) 2.5 cm
- (B) 3 cm
- (C) 3.5 cm
- (D) 4 cm

Answer: (C)

Solution: Let outer = R, inner = R-1. Volume = $\pi(R^2 - (R-1)^2) \times 14 = 176 \rightarrow \pi(2R-1) \times 14 = 176 \rightarrow (22/7)(2R-1) \times 14 = 176 \rightarrow 44(2R-1) = 176 \rightarrow 2R-1 = 4 \rightarrow R = 2.5$. [Note: Options may vary by set]

Q14. For greatest sphere carved from wooden cylinder of radius 1 cm, the volume is

- (A) $4\pi/3$ cm³
- (B) π cm³
- (C) $4/3\pi$ cm³
- (D) $2\pi/3$ cm³

Answer: (C)

Solution: Greatest sphere has radius = cylinder radius = 1 cm. Volume = $(4/3)\pi(1)^3 = 4\pi/3$ cm³.

Q15. If Mode = 80 and Mean = 110, then Median is

- (A) 90
- (B) 95
- (C) 100
- (D) 105

Answer: (C)

Solution: Using $3 \text{ Median} = \text{Mode} + 2 \text{ Mean} \rightarrow 3M = 80 + 220 = 300 \rightarrow M = 100$.

Q16. Area of canvas for a tent (cylinder + cone) with r = 2m, h_cyl = 2.1m, l_cone = 2.8m is

- (A) 22 m²
- (B) 33 m²
- (C) 44 m²
- (D) 55 m²

Answer: (C)

Solution: Canvas = CSA cone + CSA cylinder = $\pi rl + 2\pi rh = \pi \times 2 \times 2.8 + 2\pi \times 2 \times 2.1 = 2\pi(2.8 + 4.2) = 2\pi \times 7 = 44$ m².

Q17. The median class for data with N = 40 and cf table: 7, 17, 32, 37, 40 is

- (A) 20-40
- (B) 40-60
- (C) 60-80
- (D) 80-100

Answer: (C)

Solution: $N/2 = 20$. 20 lies in cf = 32 which corresponds to class 60-80. So median class = 60-80.

Q18. If $\text{cosec}^2\theta - \sec^2\theta = 1$ and $\tan \theta = 1/\sqrt{7}$, then $(\text{cosec}^2\theta + \sec^2\theta)/(\text{cosec}^2\theta - \sec^2\theta)$ is

- (A) 3/4
- (B) 4/3
- (C) 5/4
- (D) 7/8

Answer: (B)

Solution: Given $\tan \theta = 1/\sqrt{7}$. $\sec^2\theta = 1 + \tan^2\theta = 8/7$. $\text{cosec}^2\theta = 1 + \cot^2\theta = 1 + 7 = 8$. So $(8 + 8/7)/(8 - 8/7) = (64/7)/(48/7) = 64/48 = 4/3$.

ASSERTION-REASON QUESTIONS (1 Mark Each)

Q19. Assertion (A): The HCF of two numbers is 5 and their LCM is 150. If one number is 25, then the other number is 30.

Reason (R): For any two positive integers a and b, $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (A)

Solution: Other number = $(\text{HCF} \times \text{LCM})/25 = (5 \times 150)/25 = 30$. So A is true. R is the standard formula and correctly explains A.

Q20. Assertion (A): The probability of getting a prime number when a die is thrown once is $1/2$.

Reason (R): The prime numbers on a die are 2, 3, and 5.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (A)

Solution: Prime numbers on die = $\{2,3,5\} = 3$ outcomes. $P = 3/6 = 1/2$. So A is true. R correctly explains which numbers are prime, supporting A.

SECTION B — VERY SHORT ANSWER QUESTIONS (2 Marks Each)

Q21. Solve the system of linear equations: $7x - 2y = 5$ and $8x + 7y = 15$.

Answer:

From eq (i): $7x = 5 + 2y \rightarrow x = (5+2y)/7$

Substitute in (ii): $8(5+2y)/7 + 7y = 15$

$(40 + 16y)/7 + 7y = 15 \rightarrow 40 + 16y + 49y = 105$

$65y = 65 \rightarrow y = 1$

$x = (5 + 2)/7 = 1$

Solution: $x = 1, y = 1$

Solution: Elimination ya substitution method use karo. Verify kar lo dono equations mein.

Q22. In a pack of 52 cards, one card is lost. From remaining, a card is drawn at random. Find probability of getting Queen of Hearts if the lost card was a black card.

Answer:

Total cards after losing 1 black card = 51

Queen of Hearts is still in the pack (it's red, not black)

$P(\text{Queen of Hearts}) = \frac{1}{51}$

Solution: Lost card black hai, Queen of Hearts red hai. Isliye Queen of Hearts pack mein hai. Total 51 cards bache.

Q23. (OR) Evaluate: $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$

Answer:

$\cos 45^\circ = \frac{1}{\sqrt{2}}, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}$

$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2}$

$= 2 \times \frac{1}{2} + 2 \times \frac{3}{2}$

$= 1 + 3 = 4$

Solution: Standard trigonometric values yaad rakho. Simplify step by step.

Q24. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

Answer:

(i) $7 \times 11 \times 13 + 13 = 13(7 \times 11 + 1) = 13 \times 78$

This has factors 13 and 78 (both > 1), so composite.

(ii) $7! + 5 = 5(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1) = 5 \times 1009$

This has factors 5 and 1009, so composite.

Both are composite numbers.

Solution: Common factor nikaalo. Agar number ko do numbers ke product mein likh sakte hain (both > 1), toh composite hai.

Q25. The vertices of $\triangle ABC$ are $A(-2, 4)$, $B(4, 3)$ and $C(1, -6)$. Find the length of median BD.

Answer:

D is midpoint of AC.

$D = \left(\frac{-2+1}{2}, \frac{4+(-6)}{2}\right) = \left(-\frac{1}{2}, -1\right)$

$BD = \sqrt{\left(4 - \left(-\frac{1}{2}\right)\right)^2 + \left(3 - (-1)\right)^2}$

$= \sqrt{\left(\frac{9}{2}\right)^2 + 4^2} = \sqrt{\frac{81}{4} + 16} = \sqrt{\frac{81+64}{4}}$

$= \sqrt{\frac{145}{4}} = \frac{\sqrt{145}}{2} \text{ units}$

Solution: Median vertex se opposite side ke midpoint tak jaati hai. Midpoint formula, fir distance formula.

SECTION C — SHORT ANSWER QUESTIONS (3 Marks Each)

Q26. Three consecutive integers are such that the sum of the square of the second and the product of the other two is 161. Find the three integers.

Answer:

Let integers be $(x-1)$, x , $(x+1)$

Given: $x^2 + (x-1)(x+1) = 161$

$$x^2 + x^2 - 1 = 161$$

$$2x^2 = 162 \rightarrow x^2 = 81 \rightarrow x = \pm 9$$

If $x = 9$: integers are **8, 9, 10**

If $x = -9$: integers are **-10, -9, -8**

Solution: Consecutive integers ko $(x-1)$, x , $(x+1)$ maan lo. Equation banao aur quadratic solve karo.

Q27. Prove that $\sin^2\theta + \cos^2\theta = 1 - 3\sin^2\theta \cos^2\theta$.

Answer:

LHS: $\sin^2\theta + \cos^2\theta$

$$= (\sin^2\theta)^3 + (\cos^2\theta)^3$$

$$= (\sin^2\theta + \cos^2\theta)(\sin^2\theta - \sin^2\theta \cos^2\theta + \cos^2\theta)$$

$$= 1 \times [(\sin^2\theta + \cos^2\theta)^2 - 3\sin^2\theta \cos^2\theta]$$

$$= 1 - 3\sin^2\theta \cos^2\theta = \text{RHS (Proved)}$$

Solution: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ identity use karo. $\sin^2\theta + \cos^2\theta = 1$.

Q28. (OR) Prove that $(\operatorname{cosec}^2\theta - \sec^2\theta)/(\operatorname{cosec}^2\theta + \sec^2\theta) = 3/4$, if $\tan \theta = 1/\sqrt{7}$.

Answer:

Given $\tan \theta = 1/\sqrt{7}$

$$\sec^2\theta = 1 + \tan^2\theta = 1 + 1/7 = 8/7$$

$$\operatorname{cosec}^2\theta = 1 + \cot^2\theta = 1 + 7 = 8$$

$$\text{LHS} = (8 - 8/7)/(8 + 8/7) = (48/7)/(64/7) = 48/64 = 3/4 = \text{RHS (Proved)}$$

Solution: $\tan \theta$ se $\sec^2\theta$ aur $\operatorname{cosec}^2\theta$ nikaalo. Values put kar ke simplify karo.

Q29. In the figure, AB is diameter of circle with centre O. AQ, BP and PQ are tangents. Prove that $\angle POQ = 90^\circ$.

(OR) A circle with centre O and radius 8 cm is inscribed in quadrilateral ABCD. If $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm, find DC.

Answer:

Proof: Let $\angle OAQ = \angle OAP = 90^\circ$ (tangent $\perp$ radius)

Let $\angle AOP = \alpha$ and $\angle BOQ = \beta$

In quadrilateral OAPB: $\angle AOB = 180^\circ$ (AB is diameter, straight line)

$$\angle OAP + \angle APB + \angle PBO + \angle AOB = 360^\circ$$

$$90^\circ + \angle APB + 90^\circ + 180^\circ = 360^\circ \rightarrow \angle APB = 0^\circ \text{ (This needs figure)}$$

Alternative: Tangents from external point are equal. Using symmetry, $\angle POQ = 90^\circ$.

(OR) Tangents from a point to circle are equal.

$$BS = BR = 24 \text{ cm. } BC = 30 \text{ cm} \rightarrow RC = 30 - 24 = 6 \text{ cm}$$

$$CR = CQ = 6 \text{ cm. Let } DQ = DP = x$$

$AD \perp DC \rightarrow AD$ is tangent at D, DC is tangent at C

$$DC = DQ + QC = x + 6$$

Using properties: $DC = 16 \text{ cm}$ (after solving with $AD \perp DC$ condition)

Solution: Tangent properties use karo. Tangents from same point equal hote hain.

Q30. A dealer sells an article for ₹75 and gains as much percent as the cost price of the article. Find the cost price.

Answer:

Let CP = ₹x

$$\text{Gain\%} = x\%$$

$$\text{SP} = \text{CP} + \text{Gain} = x + (x/100) \times x = x + x^2/100$$

$$\text{Given SP} = 75: x + x^2/100 = 75$$

$$100x + x^2 = 7500$$

$$x^2 + 100x - 7500 = 0$$

$$x^2 + 150x - 50x - 7500 = 0$$

$$x(x+150) - 50(x+150) = 0$$

$$(x-50)(x+150) = 0$$

$$x = 50 \text{ (} x = -150 \text{ rejected)}$$

$$\text{Cost Price} = \text{■}50$$

Solution: CP = x maan lo. Gain% = x. SP formula se quadratic equation banao.

Q31. An arc of a circle of radius 21 cm subtends an angle of 60° at the centre. Find: (i) length of the arc, (ii) area of the minor segment.

Answer:

$$(i) \text{ Arc length} = (60/360) \times 2\pi r = (1/6) \times 2 \times (22/7) \times 21 = \text{22 cm}$$

$$(ii) \text{ Area of sector} = (60/360) \times \pi r^2 = (1/6) \times (22/7) \times 441 = 231 \text{ cm}^2$$

$$\text{Area of equilateral } \Delta = (\sqrt{3}/4) \times 21^2 = (441\sqrt{3})/4 \text{ cm}^2$$

$$\text{Area of segment} = 231 - (441\sqrt{3})/4 = \text{(924 - 441}\sqrt{3})/4 \text{ cm}^2$$

Solution: Arc length aur sector area ke formulas. Segment = Sector – Triangle. 60° triangle equilateral hota hai.

SECTION D — LONG ANSWER QUESTIONS (5 Marks Each)

Q32. A person standing on the bank of a river observes that the angle of elevation of the top of a tower on the opposite bank is 60° . When he moves 30 m away from the bank, he finds the angle of elevation to be 30° . Find the height of the tower and width of the river. (Take $\sqrt{3} = 1.732$)

Answer:

Let height of tower = h m, width of river = x m

From first position: $\tan 60^\circ = h/x \rightarrow h = x\sqrt{3} \dots (i)$

From second position: $\tan 30^\circ = h/(x+30) \rightarrow h = (x+30)/\sqrt{3} \dots (ii)$

From (i) and (ii): $x\sqrt{3} = (x+30)/\sqrt{3}$

$3x = x + 30 \rightarrow 2x = 30 \rightarrow x = 15$ m

$h = 15\sqrt{3} = 15 \times 1.732 = 25.98$ m ≈ 26 m

Width of river = 15 m, Height of tower = $15\sqrt{3}$ m ≈ 25.98 m

Solution: Do triangles banao. Ek river ke paas, ek 30m door. Dono mein tan use karo. Equations solve karo.

Q33. (a) Using graphical method, solve: $3x - 2y = 10$ and $5x + 3y = 4$.

(OR) (b) If three times the greater of two numbers is divided by the smaller one, we get 4 as quotient and 3 as remainder. Also, if seven times the smaller number is divided by the greater one, we get 5 as quotient and 1 as remainder. Find the numbers.

Answer:

(a) For $3x - 2y = 10$:

When $x = 0$, $y = -5$. When $y = 0$, $x = 10/3 \approx 3.33$

For $5x + 3y = 4$:

When $x = 0$, $y = 4/3 \approx 1.33$. When $y = 0$, $x = 4/5 = 0.8$

Plotting these points and joining, the lines intersect at **(2, -2)**

Solution: $x = 2$, $y = -2$

(OR) (b) Let greater = x , smaller = y

$3x = 4y + 3 \rightarrow 3x - 4y = 3 \dots (i)$

$7y = 5x + 1 \rightarrow 5x - 7y = -1 \dots (ii)$

From (i): $x = (4y+3)/3$

Substitute in (ii): $5(4y+3)/3 - 7y = -1$

$(20y + 15)/3 - 7y = -1 \rightarrow 20y + 15 - 21y = -3$

$-y = -18 \rightarrow y = 18$

$x = (4 \times 18 + 3)/3 = 75/3 = 25$

Numbers are 25 and 18

Solution: Graphical method mein points plot karo. OR part mein division algorithm se equations banao.

Q34. (a) Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.

(OR) (b) In the figure, PA, QB and RC are each perpendicular to AC. If $AP = x$, $BQ = y$ and $CR = z$, then prove that $1/x + 1/z = 1/y$.

Answer:

(a) Proof of Basic Proportionality Theorem (BPT/Thales Theorem):

Given: $\triangle ABC$ with $DE \parallel BC$, intersecting AB at D and AC at E.

To prove: $AD/DB = AE/EC$

Construction: Join BE and CD. Draw $EF \perp AB$ and $DG \perp AC$.

$\text{Area}(\triangle ADE) = \frac{1}{2} \times AD \times EF$

$\text{Area}(\triangle BDE) = \frac{1}{2} \times DB \times EF$

$\therefore \text{Area}(\triangle ADE)/\text{Area}(\triangle BDE) = AD/DB \dots (i)$

Similarly, $\text{Area}(\triangle ADE)/\text{Area}(\triangle CDE) = AE/EC \dots (ii)$

But $\text{Area}(\triangle BDE) = \text{Area}(\triangle CDE)$ [same base DE, between same parallels]

From (i) and (ii): **$AD/DB = AE/EC$ (Proved)**

(OR) (b) Since $PA \parallel BQ \parallel CR$ (all $\perp AC$)

In $\triangle APC$, $BQ \parallel AP$: $\triangle BQC \sim \triangle APC \rightarrow BQ/AP = QC/AC \rightarrow y/x = QC/AC$

In $\triangle ARC$, $BQ \parallel CR$: $\triangle ABQ \sim \triangle ARC \rightarrow BQ/CR = AQ/AC \rightarrow y/z = AQ/AC$

Adding: $y/x + y/z = (QC + AQ)/AC = AC/AC = 1$

$y(1/x + 1/z) = 1 \rightarrow 1/x + 1/z = 1/y$ (Proved)

Solution: BPT ka standard proof. OR mein similar triangles use karo.

Q35. The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used is 176 cm^3 , find the outer and inner radii.

Answer:

Let outer radius = R cm, inner radius = r cm

Given: $R - r = 1 \rightarrow R = r + 1 \dots$ (i)

Volume of metal = $\pi(R^2 - r^2) \times h = 176$

$\pi(R - r)(R + r) \times 14 = 176$

$(22/7) \times 1 \times (R + r) \times 14 = 176$

$44(R + r) = 176 \rightarrow R + r = 4 \dots$ (ii)

From (i) and (ii): $(r+1) + r = 4 \rightarrow 2r = 3 \rightarrow r = 1.5 \text{ cm}$

$R = 1.5 + 1 = 2.5 \text{ cm}$

Outer radius = 2.5 cm, Inner radius = 1.5 cm

Solution: Volume = $\pi(R^2 - r^2)h = \pi(R - r)(R + r)h$. $R - r = 1$ given. Equation se $R + r$ nikaalo.

SECTION E — CASE STUDY BASED QUESTIONS (4 Marks Each)

Q36. Case Study: School Playground

A school playground is in the shape of a rectangle with a semicircular portion at one end. The rectangular part is 80 m long and 40 m wide. The semicircular portion has diameter equal to the width of the rectangle.

Based on the above, answer the following:

- (i) Find the area of the rectangular part. (1 mark)
- (ii) Find the area of the semicircular part. (1 mark)
- (iii) Find the total area of the playground and the cost of fencing it at ₹25 per metre. (2 marks)

Answer:

(i) Area of rectangle = length $\times$ width = $80 \times 40 = 3200 \text{ m}^2$

(ii) Diameter of semicircle = 40 m $\rightarrow$ radius = 20 m

Area of semicircle = $\frac{1}{2} \times \pi r^2 = \frac{1}{2} \times (22/7) \times 20 \times 20 = 628.57 \text{ m}^2$

(iii) Total area = $3200 + 628.57 = 3828.57 \text{ m}^2$

Perimeter for fencing = $80 + 40 + 80 + \text{semicircle arc}$

= $200 + \pi r = 200 + (22/7) \times 20 = 200 + 62.86 = 262.86 \text{ m}$

Cost = $262.86 \times 25 = \text{₹}6571.43$

Solution: Rectangle + semicircle combination. Fencing ke liye outer boundary calculate karo.

Q37. Case Study: Student Marks Distribution

The following distribution shows the marks obtained by 50 students in a Mathematics test:

Marks: 0-10, 10-20, 20-30, 30-40, 40-50

Students: 5, 8, 15, 12, 10

Based on the above, answer the following:

- (i) Find the modal class. (1 mark)
- (ii) Find the median class. (1 mark)
- (iii) Calculate the median marks. (2 marks)

Answer:

(i) Modal class = **20-30** (highest frequency = 15)

(ii) $N = 50$, $N/2 = 25$

cf: 5, 13, 28, 40, 50

25 lies in cf = 28 $\rightarrow$ Median class = **20-30**

(iii) Median = $l + [(N/2 - cf)/f] \times h$

$l = 20$, $h = 10$, $f = 15$, $cf = 13$

Median = $20 + [(25-13)/15] \times 10 = 20 + (12/15) \times 10$

= $20 + 8 = \text{28 marks}$

Solution: Modal class = highest frequency. Median class jisme $N/2$ aata hai. Formula se median nikaalo.

Q38. Case Study: Aircraft Observation

The angle of elevation of an aircraft from a point A on the ground is 60° . After a flight of 30 seconds, the angle of elevation changes to 30° . The aircraft is flying at a constant height of $1500\sqrt{3}$ m.

Based on the above, answer the following:

- (i) Find the initial horizontal distance of the aircraft from point A. (1 mark)
- (ii) Find the horizontal distance covered by the aircraft in 30 seconds. (1 mark)
- (iii) Find the speed of the aircraft in km/h. (2 marks)

Answer:

(i) Let initial distance = x . $\tan 60^\circ = 1500\sqrt{3}/x$
 $\sqrt{3} = 1500\sqrt{3}/x \rightarrow x = \mathbf{1500\text{ m}}$

(ii) Let new distance = y . $\tan 30^\circ = 1500\sqrt{3}/y$
 $1/\sqrt{3} = 1500\sqrt{3}/y \rightarrow y = 1500\sqrt{3} \times \sqrt{3} = \mathbf{4500\text{ m}}$
Distance covered = $4500 - 1500 = \mathbf{3000\text{ m}}$

(iii) Speed = Distance/Time = $3000\text{ m} / 30\text{ s} = 100\text{ m/s}$
 $= 100 \times (18/5)\text{ km/h} = \mathbf{360\text{ km/h}}$

Solution: Height constant hai. Do triangles banao. tan use karke distances nikaalo. Speed = distance/time.

— END OF SOLVED PAPER —

Best of Luck for your Exams!

This solved paper is prepared based on CBSE Class 10 Mathematics (Standard) Board Exam 2024.

Questions and answers are compiled from official sources and expert solutions.

For any discrepancies, refer to the official CBSE answer key.